Find a Real Problem

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

One connected problem-framing brief.

Frame the activity need, challenge unsupported claims and complete the independent box brief before auditing coverage.

Keep one habit: Keep observation, hypothesis, proposed solution and later evaluation distinct.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Humans Set Boundaries. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Prompts Are Design Instructions. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Fair for Whom?. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ Earlier wording, alternatives and picture transcripts

Humans Set Boundaries

Prompts Are Design Instructions

Fair for Whom?

W1 · Frame the observed activity problem

> Use the supplied ten-request reference rather than a technology-first wish.

Your evidence record

- Activity-project reference for W1–W3: ten fictional requests used an activity catalogue. Six first choices fit the stated time and room conditions; four did not and required another choice. The log does not establish the cause, every pupil's feelings or the benefit of an unbuilt tool. The permitted project is a paper process using fictional requests and an approved catalogue; necessary request fields are available minutes and quiet-room requirement. No names, profiles, purchases or public posting are needed.

- Use the ten-request activity log to write a **problem statement** with a **user**, setting and observed difficulty. Explain why "Build an AI app" is not yet a **problem statement**.

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Separate findings from unsupported conclusions

> Keep feelings, causes and future benefit distinct from the log.

Your evidence record

- A team says that the four mismatches prove every pupil dislikes the catalogue and that a recommender will fix the problem. Identify two unsupported conclusions and state what the log actually establishes.

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Choose scope, inputs and a simpler alternative

> Define compatible results and preserve user choice.

Your evidence record

- The paper project must suggest an activity compatible with available time and a quiet-room requirement. Choose relevant **input**s from minutes available, quiet required, pupil name and home address.

  State the **output** and a simpler alternative to a recommender. Explain one boundary.

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Frame a different problem independently

> Keep one first attempt before hints or feedback.

Read and explain independently

In a fictional equipment cupboard exercise, eight teams choose a box from a list. Three boxes do not contain the items required by the corresponding request; five do.

The teacher permits a paper box-finder using fictional item requests and an inventory. Write a brief naming the **user** difficulty, **evidence**, desired **outcome**, necessary information, an excluded action and a simple alternative.

Identify one cause or benefit that these counts do not establish.

> **My independent explanation, trace and limits**

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Frame a different problem independently

MAIN LESSON · SECTION 16

W5 · Audit an always-no-match box finder

> After W4, inspect useful coverage as well as wrong selections.

Your evidence record

- For W4, a proposed box-finder never selects a wrong box because it always reports no match. Explain why counting only wrong suggestions would hide a failure.

  Name two results to record on test requests and explain why the original three-of-eight **observation** is not already an evaluation of the new tool.

> **My reasoning and evidence**

MAIN LESSON · SECTION 9

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

Choose the Classroom Sorter's real **user** and measurable **need**; keep an AI-free solution option open.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Identify claims beyond the **evidence** for A problem worth solving. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for A problem worth solving. Proposed explanation: “Using AI is itself the **outcome**”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if **need**ed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

Students spend three minutes finding returned books. Compare shelf labels with an AI locator. Define one fair success test before choosing a solution.

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

Shared markers are often lost because tray labels are unclear.

State the **need** before selecting AI.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

Learners often cannot find borrowed classroom books. Define the problem before choosing an AI tool. Suggest one simple solution to compare with AI.

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 29 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Reference · Original target response space**

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

Students spend three minutes finding returned books. Compare shelf labels with an AI locator. Define one fair success test before choosing a solution.

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · Independent-case response copy

REFERENCE · SECTION 15

Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

Use this when helpful

In a fictional equipment cupboard exercise, eight teams choose a box from a list. Three boxes do not contain the items required by the corresponding request; five do.

The teacher permits a paper box-finder using fictional item requests and an inventory. Write a brief naming the **user** difficulty, **evidence**, desired **outcome**, necessary information, an excluded action and a simple alternative.

Identify one cause or benefit that these counts do not establish.

My reasoning and evidence

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

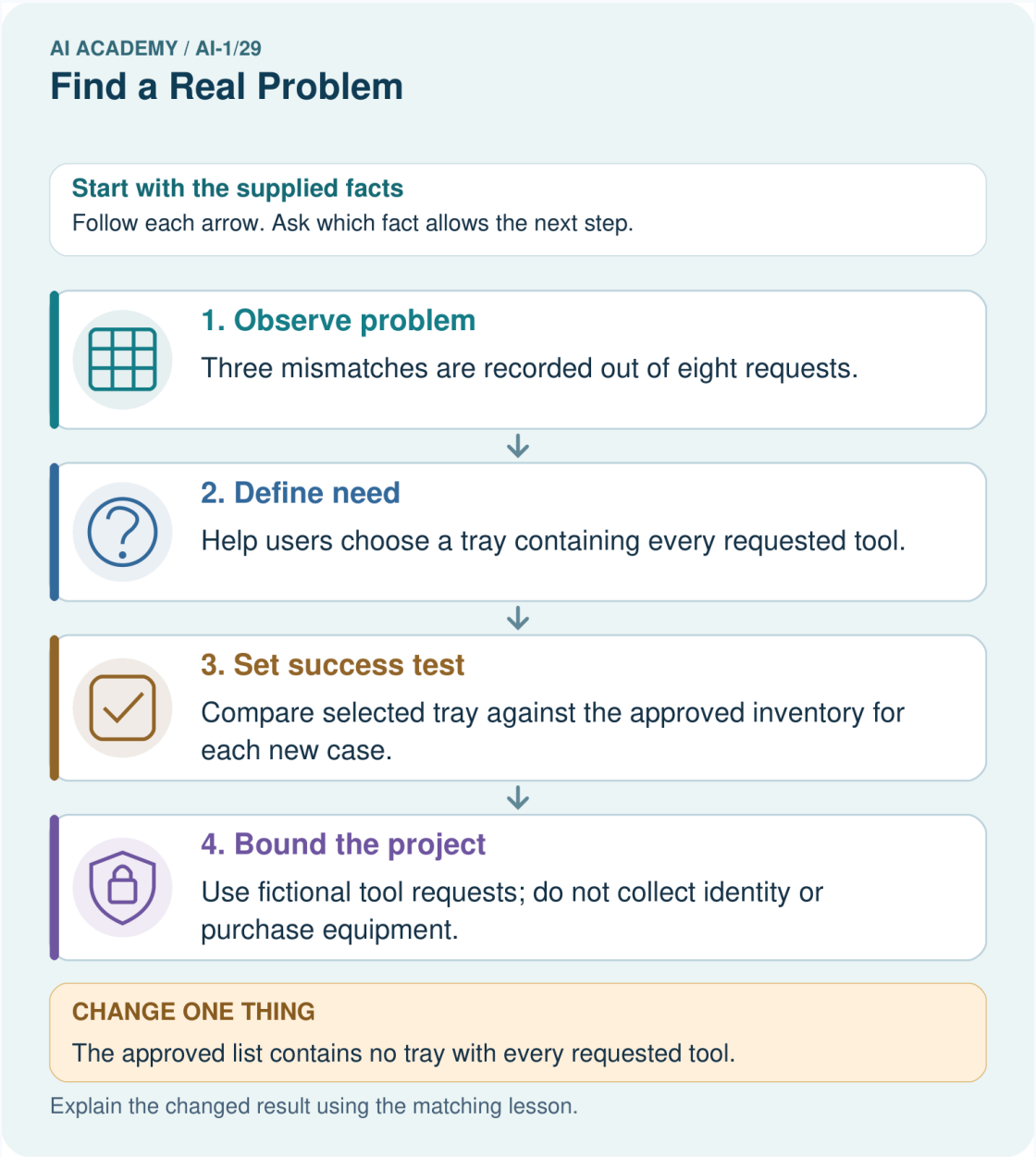
> Keep the explanation and the support you used.

Use this when helpful

A fictional class loses time finding the correct craft tray.

In eight recorded requests, three selected trays lack a required tool; five match.

The teacher permits a paper finder using fictional requests and an approved tray list, with no pupil names.



Follow the labelled picture with the matching explanation. Open the vector diagram

Write a one-sentence project goal and say what **evidence** would count as a failure.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For the Helpful Classroom Sorter, retain the problem record, permitted scope, baseline and observable acceptance criteria.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Helpful Classroom Sorter

Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

For the Helpful Classroom Sorter, retain the problem record, permitted scope, baseline and observable acceptance criteria.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional class loses time finding the correct craft tray.
- Which labelled step matches this detail? Use fictional tool requests; do not collect identity or purchase equipment.
- 1. Bound the project
- 2. Observe problem

- 3. Set success test
- 4. Define **need**
- Hint if **need**ed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Bound the project Use fictional tool requests; do not collect identity or purchase equipment. The mismatch record describes a concrete **need**.

A measurable goal checks whether the chosen tray satisfies the request; calling the system smart would not tell us how to evaluate it.

Start with a transparent paper baseline, and keep unsatisfied requests visible instead of selecting a near match without warning.

Changed-case question

- Changed-case practice: The approved list contains no tray with every requested tool. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. The finder should report no suitable tray rather than guess.
- 3. This one example guarantees correct results for every future **input**.

- Hint if **need**ed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> ---
>
> ---
>
> ---
>
> ---

▼ **Check the changed case**

After your attempt, compare with the supported explanation: The finder should report no suitable tray rather than guess. Success includes honestly identifying unsatisfied requirements.

Repair and save your work

Review Humans Set Boundaries (ai-1/28). Goal: Distinguish a system's capability, the correctness of its **output** and permission to take an action. Short practice: Which original proposal is authorised?

Why do the other two **need** separate approval?. Return to this week and explain what you changed.

Review Prompts Are Design Instructions (ai-1/26). Goal: Write a prompt with a clear task, supplied **evidence** and checkable **output** requirements.

Short practice: What must happen if B instead conflicts with another equally authoritative duration record?. Return to this week and explain what you changed.

Review Fair for Whom? (ai-1/23). Goal: Compute a correct-out-of-assessed fraction and explain its denominator. Short practice: Which group loses performance under N, and can total improvement erase that fact?.

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

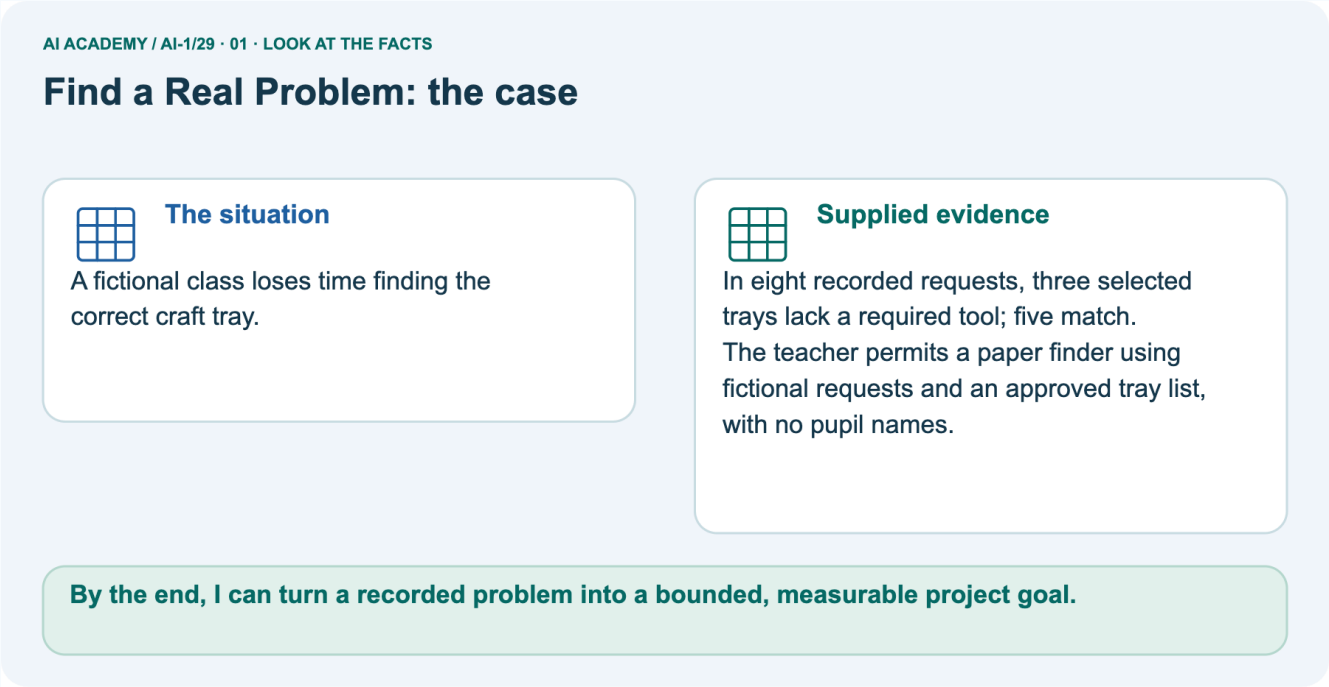
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

Choose the image that helps rather than repeating all six.

01. Find a Real Problem: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

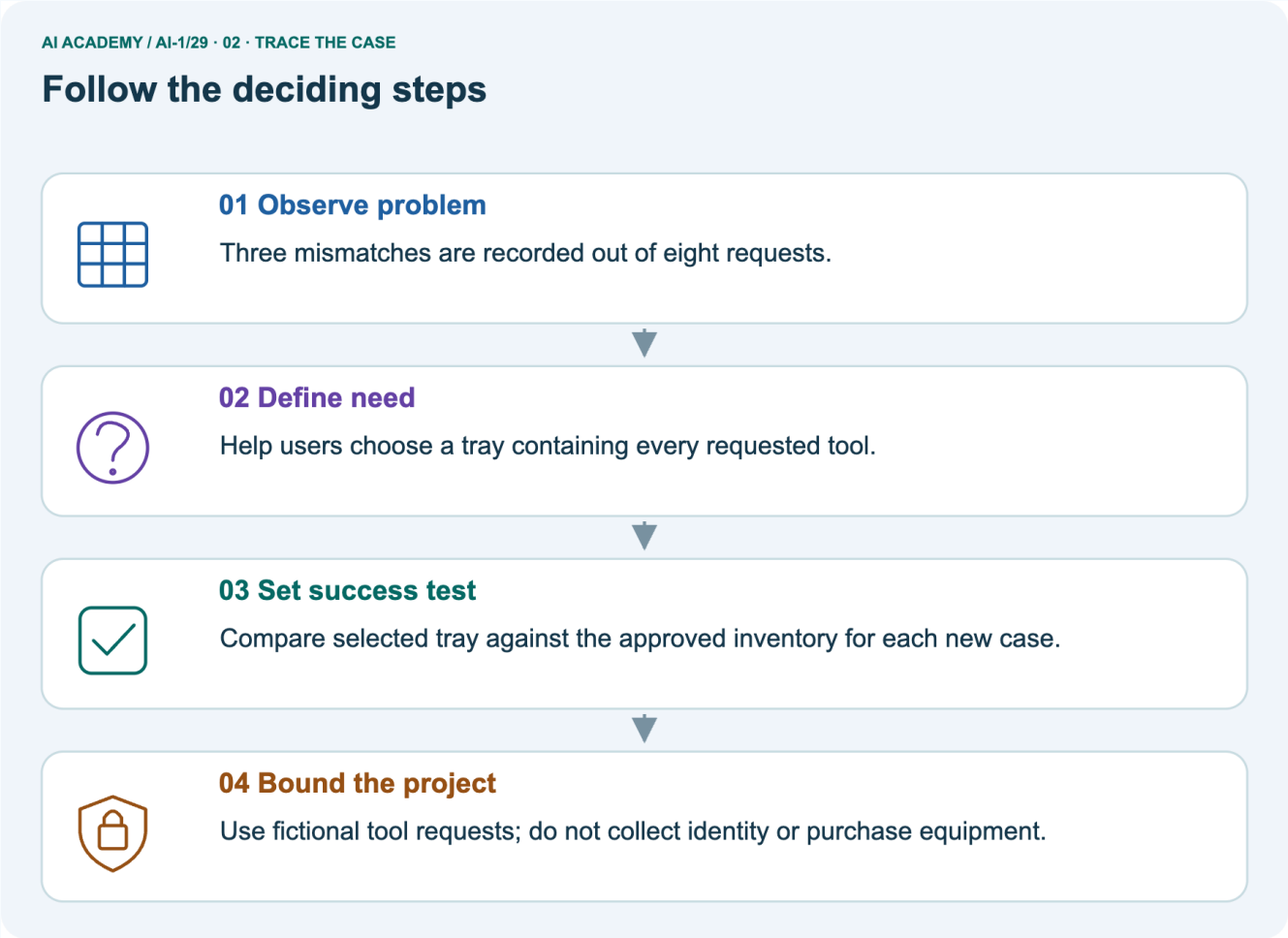
A fictional class loses time finding the correct craft tray. Supplied facts: In eight recorded requests, three selected trays lack a required tool; five match.

The teacher permits a paper finder using fictional requests and an approved tray list, with no pupil names.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Observe problem: Three mismatches are recorded out of eight requests. → Define **need**: Help **user**s choose a tray containing every requested tool. → Set success test: Compare selected tray against the approved inventory for each new case. → Bound the project: Use fictional tool requests; do not collect identity or purchase equipment.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. From an observed difficulty to a testable goal



Follow the labelled picture with the matching explanation. Open the vector diagram

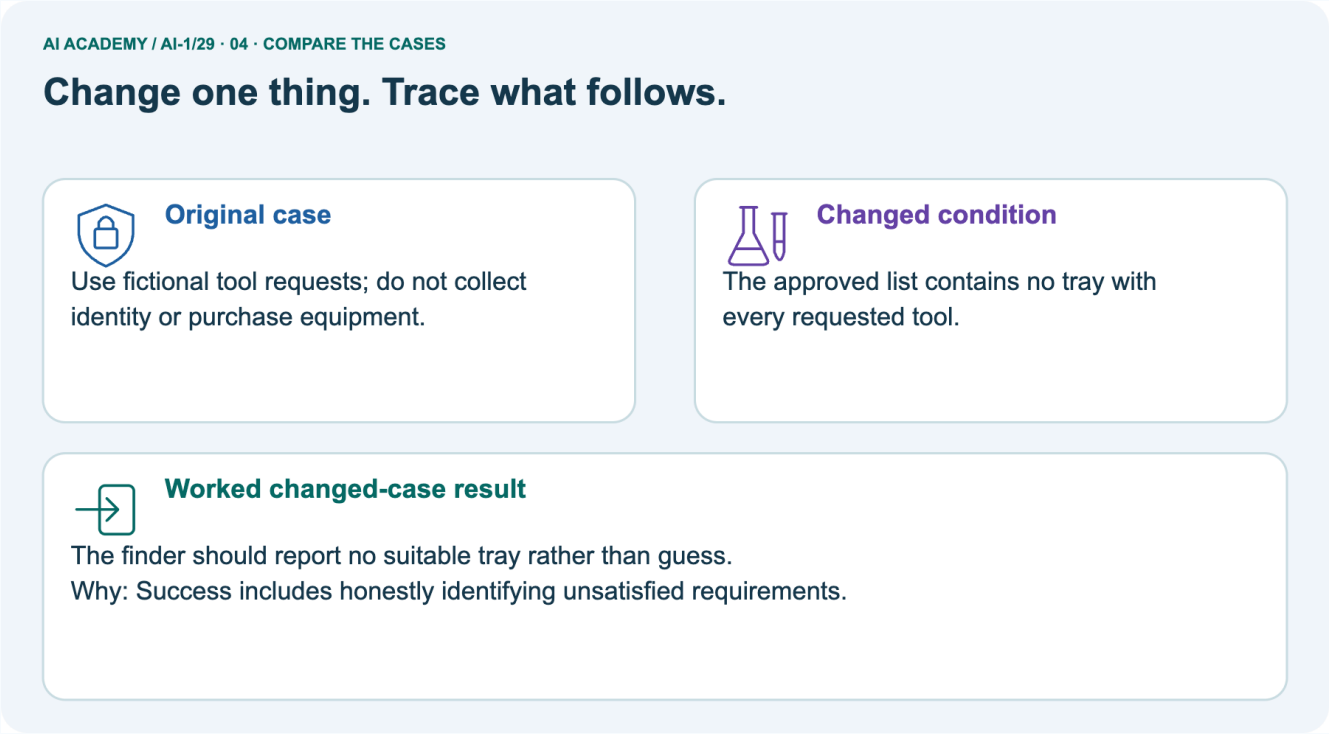
From an observed difficulty to a testable goal. 4 of 10 choices mismatch | Some re-selection occurred | Cause is not established. **Need** a compatible choice | Check time and conditions | **User** can still decline.

Paper suggestion **process** | Small permitted prototype | Benefit **need**s a comparison. Measure fit and coverage | Observe useful **outcome**s | Do not hide no-result cases

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

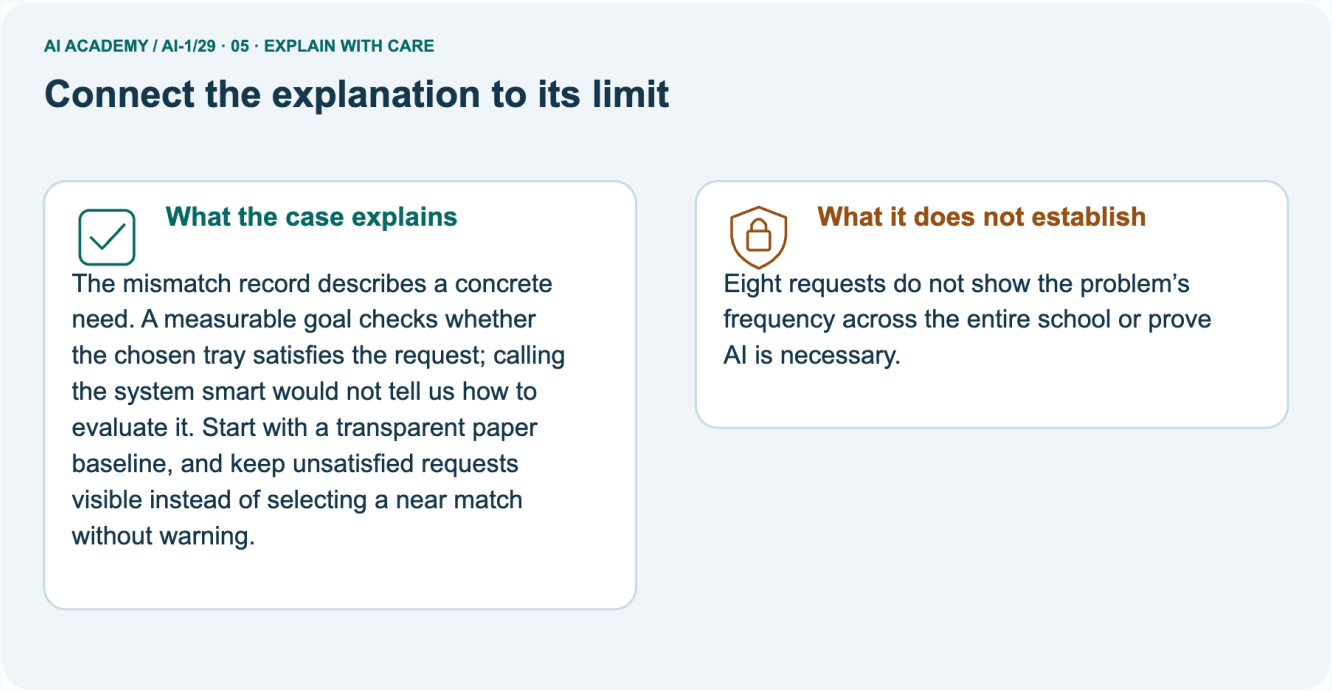
Original case: Use fictional tool requests; do not collect identity or purchase equipment.
Changed condition: The approved list contains no tray with every requested tool.

Expected result: The finder should report no suitable tray rather than guess. Reason: Success includes honestly identifying unsatisfied requirements.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

The mismatch record describes a concrete **need**. A measurable goal checks whether the chosen tray satisfies the request; calling the system smart would not tell us how to evaluate it.

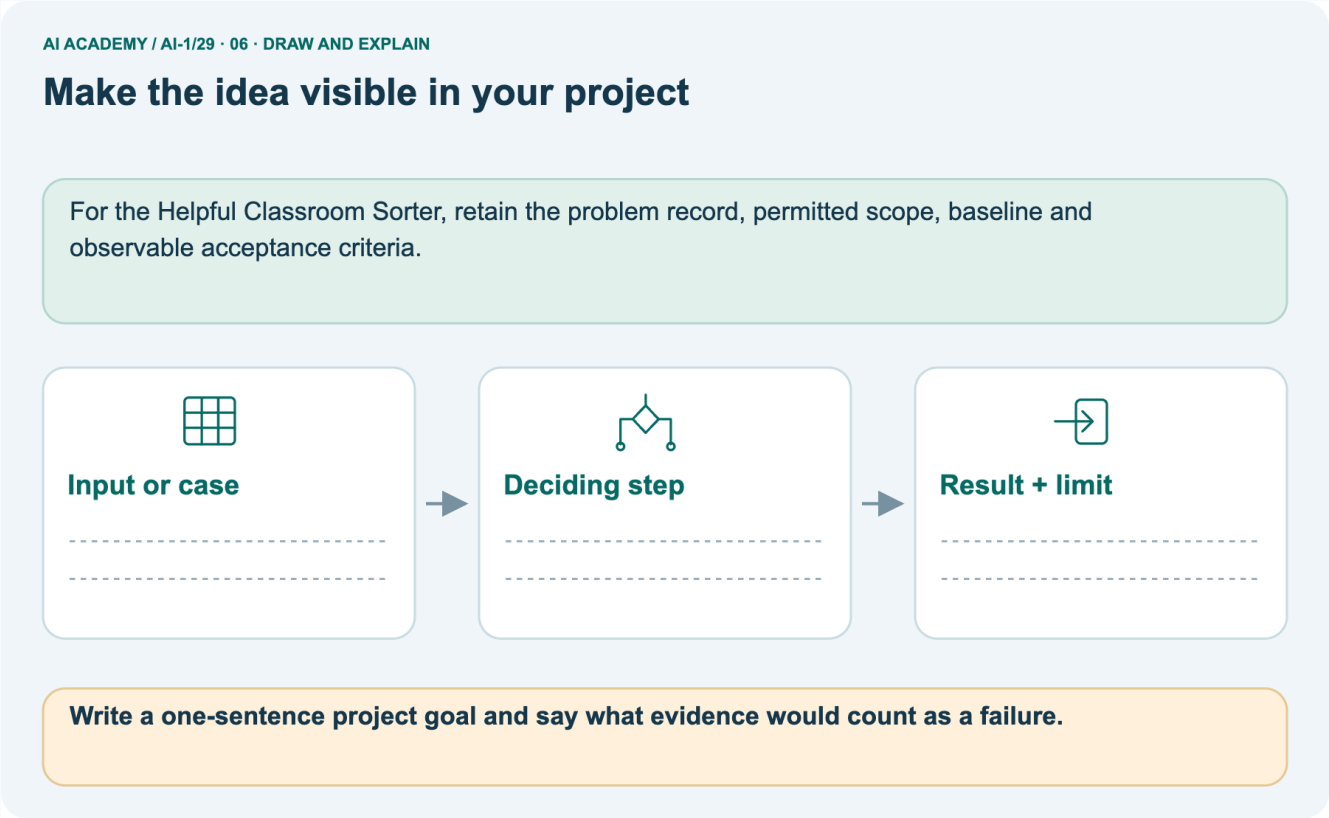
Start with a transparent paper baseline, and keep unsatisfied requests visible instead of selecting a near match without warning.

Limit: Eight requests do not show the problem's frequency across the entire school or prove AI is necessary.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Helpful Classroom Sorter, retain the problem record, permitted scope, baseline and observable acceptance criteria. Draw your **input**, deciding step, result and limit.

Supported question: Write a one-sentence project goal and say what **evidence** would count as a failure.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Find a Real Problem: the case

02. Follow the deciding steps

03. From an observed difficulty to a testable goal

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.